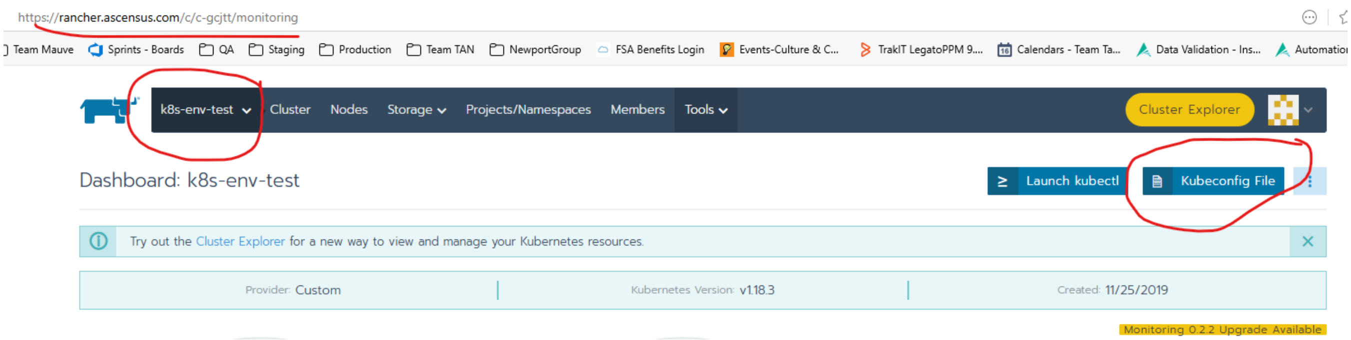
Blazemeter with Kubernetes & Helm

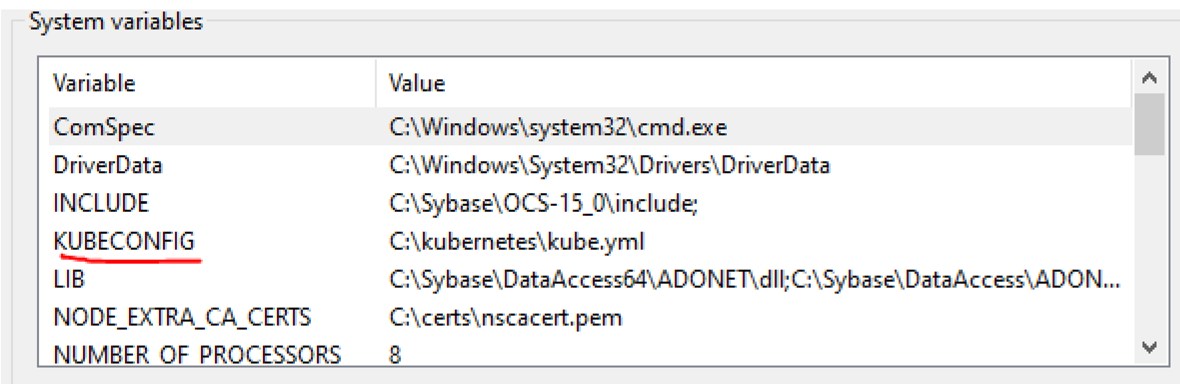
This document is a work in progress. For now it includes the commands used for the government savings demo with Dong Do

Installing Helm and KubeCtl

1. Download helm windows (https://github.com/helm/helm/releases)
    a. Click here for Installing Helm
2. Download **KubeCTL** windows (https://kubernetes.io/releases/download/#binaries)
3. Add both exe to a folder and make sure that folder is in the windows **Path** in your **Environment Variables**
4. Next step is to download the **kube.yml** from the cluster, save it as a file.
    i. To download kube.yml, go to Rancher https://rancher.ascensus.com/.
    ii. Go to **k8s-env-test.** Click on Kuberconfig file. Download or copy to clipboard from



5. Create this **System variable** in your **Environment Variables** pointing to the file


System variables

Variable	Value
ComSpec	C:\Windows\system32\cmd.exe
DriverData	C:\Windows\System32\Drivers\DriverData
INCLUDE	C:\Sybase\OCS-15_0\include;
KUBECONFIG	C:\kubernetes\kube.yml
LIB	C:\Sybase\DataAccess64\ADONET\dll;C:\Sybase\DataAccess\ADON...
NODE_EXTRA_CA_CERTS	C:\certs\nscacert.pem
NUMBER_OF_PROCESSORS	8

6. Once this is set, go to command line and run **kubectl describe nodes**. Ensure you see the same nodes as are in the cluster

Sample repo with job configuration for helm/kubectl

https://tfs.ascensus.com/tfs/PSG/Portfolio/_git/EmployeePublicApi?path=%2FTests%2FPerformanceTests%2FHelm%2Ftemplates%2Ftaurus-deployment.yaml

1. Create a **performance-test.yaml** file in the same folder where you have kubectl.exe.

performance-test.yaml

```
execution:
- concurrency: 100
  ramp-up: 1m
  hold-for: 5m
  scenario: quick-test

scenarios:
  quick-test:
    requests:
    - http://blazedemo.com
```

2. Create a **blazemeter.yaml** file in the same folder where you have kubectl.exe.

blazemeter.yaml

```
reporting:
- module: "blazemeter"
  report-name: "Testing Taurus Only"
  test: "529 Blazemeter Demo"
  project: 631844

modules:
  blazemeter:
    token: 7f7061aba632ea200b7a0596:
253b614a703a258268efdf1d8f954144355611bf4910de38ed697083598dc518277ee9f2
```

3. Create a **client-deployment.yaml** file in the same folder where you have kubectl.exe. This is the taurus side of the performance test

client-deployment.yaml

```
apiVersion: v1
kind: Pod
metadata:
  namespace: performance-testing
  name: taurus
  labels:
    app: taurus
spec:
  restartPolicy: OnFailure
  containers:
  - name: taurus
    image: proget.ascensus.com/ascensus/taurus:1.14.1
    imagePullPolicy: Always
    args:
    - performance-test.yaml
    - blazemeter.yaml
    volumeMounts:
    - name: config
      mountPath: /bzt-configs
    - name: test-artifacts
      mountPath: /bzt-configs/TestArtifacts
  volumes:
  - name: config
    configMap:
      name: performance-test-config
  - name: test-artifacts
    emptyDir: {}
```

4. Ensure **KUBECONFIG** in Environment Variable is pointed to the correct cluster. You can use a kube.yml file.
   export KUBECONFIG=$PWD/kube.yml
5. Run below commands
   a. kubectl create namespace performance-testing
   b. kubectl delete configmap performance-test-config --namespace=performance-testing
   c. kubectl create configmap performance-test-config --namespace=performance-testing --from-file performance-test.yaml --from-file blazemeter.yaml
   d. kubectl apply -f client-deployment.yaml

If the above executes correctly, you should see something like this in the logging.

```
5/28/2020 2:48:26 PM 18:48:26 INFO: Current: 100 vu 32 succ 0 fail 2.536 avg rt / Cumulative: 2.089 avg rt, 0% failures
5/28/2020 2:48:39 PM 18:48:39 WARNING: Please wait for graceful shutdown...
5/28/2020 2:48:39 PM 18:48:39 INFO: Shutting down...
5/28/2020 2:48:39 PM 18:48:39 INFO: Post-processing...
5/28/2020 2:48:41 PM 18:48:41 INFO: Changed data analysis delay to 16s
5/28/2020 2:48:41 PM 18:48:41 INFO: Test duration: 0:02:20
5/28/2020 2:48:41 PM 18:48:41 INFO: Samples count: 4355, 0.00% failures
5/28/2020 2:48:41 PM 18:48:41 INFO: Average times: total 2.195, latency 0.094, connect 0.002
5/28/2020 2:48:42 PM 18:48:42 INFO: Percentiles:
5/28/2020 2:48:42 PM +---------------+---------------+
5/28/2020 2:48:42 PM | Percentile, % | Resp. Time, s |
5/28/2020 2:48:42 PM +---------------+---------------+
5/28/2020 2:48:42 PM | 0.0 | 0.526 |
5/28/2020 2:48:42 PM | 50.0 | 1.083 |
5/28/2020 2:48:42 PM | 90.0 | 5.08 |
5/28/2020 2:48:42 PM | 95.0 | 7.804 |
5/28/2020 2:48:42 PM | 99.0 | 14.896 |
5/28/2020 2:48:42 PM | 99.9 | 25.472 |
5/28/2020 2:48:42 PM | 100.0 | 27.68 |
5/28/2020 2:48:42 PM +---------------+---------------+
5/28/2020 2:48:42 PM 18:48:42 INFO: Request label stats:
5/28/2020 2:48:42 PM +----------------------+--------+---------+--------+-------+
5/28/2020 2:48:42 PM | label | status | succ | avg_rt | error |
5/28/2020 2:48:42 PM +----------------------+--------+---------+--------+-------+
5/28/2020 2:48:42 PM | http://blazedemo.com | OK | 100.00% | 2.195 | |
5/28/2020 2:48:42 PM +----------------------+--------+---------+--------+-------+
5/28/2020 2:48:42 PM 18:48:42 INFO: Sending remaining KPI data to server...
5/28/2020 2:48:45 PM 18:48:45 INFO: Uploading all artifacts as artifacts.zip ...
5/28/2020 2:48:46 PM 18:48:46 INFO: Uploading /tmp/artifacts/bzt.log
5/28/2020 2:48:48 PM 18:48:48 INFO: Ending data feeding...
5/28/2020 2:48:48 PM 18:48:48 INFO: Online report link: https://a.blazemeter.com/app/#/masters/28789428
5/28/2020 2:48:48 PM 18:48:48 INFO: Artifacts dir: /tmp/artifacts
5/28/2020 2:48:48 PM 18:48:48 INFO: Done performing with code: 0
```

Blazemeter data will be found here

https://a.blazemeter.com/app/#/accounts/406482/workspaces/516742/projects/631844/tests/8091607